Gravity above the surface

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use distance from a planet's center to compare fields at altitude.

SUCCESS CRITERIA - I can...

1. Use r=R+h rather than altitude alone.
2. Compare fields with the inverse-square relation.
3. Explain why orbit is not a zero-gravity region.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "altitude" and "radius" to compare your model or result. Sentence starters are optional.

altitude: height above a reference surface

radius: distance from the center to the edge of a circle or sphere

field: a description of an influence at each location

01 Warm-up: bring back an earlier idea

For fixed masses, what separation multiplier makes gravity 1/9 of its original value? What multiplier is needed if both masses also become 5 times as large?

Word help: separation = distance between the specified objects or centers

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

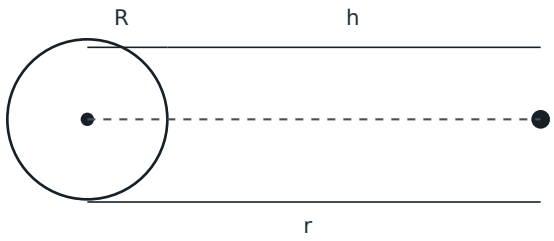
02 Worked example: explain the model

Reference: $r = R + h$; $g_h/g_{surface} = [R/(R + h)]^2$.
R is planet radius; h is altitude. g_h is field strength at altitude h.

Worked reasoning: At altitude h=2 R above a planet's surface, center distance is r=R+h=3 R. A surface field of 18 N/kg becomes 18/9=2 N/kg.

Your turn: Draw the radius and altitude as different line segments.
Mark the full center-to-object distance.

Sentence starter: Both ... . They differ in ... because ... .



Measure r from the center: r = R + h.

Marzano check - circle: ① ② ③

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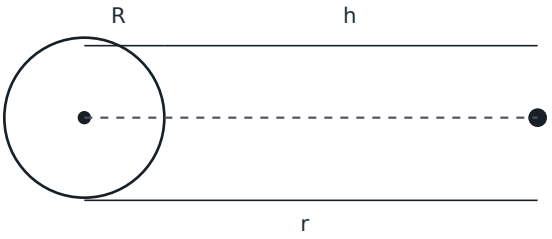
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A satellite is one planet-radius above the surface.
Label R, h and r on a sketch. What is r in units of R?

Sentence starter: My design shows … . It meets the requirement by … .



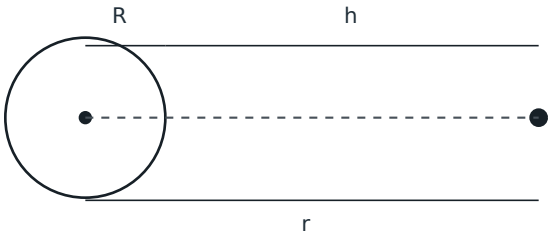
Label surface radius R, altitude h and center distance r.

Marzano check - circle: ① ② ③

04 Apply the physics

For Earth use$R=6.4\times10^{6}$ m, $g_{surface}=9.8$ N/kg.
Find g at $h=4.0\times10^{5}$ m and at h=R.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

05 Explain your reasoning

Explain why an astronaut at modest orbital altitude can float even though gravity is only slightly weaker there.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A 4 kg box has μ_s=0.50 and μ_k=0.40 on a level floor; g=9.8.
Compare pushes of 8 N and 24 N, starting from rest.
Find actual friction and acceleration.
Draw the forces on the chosen object and label their sources.

Word help: friction = a contact force opposing relative sliding or its tendency

Sentence starter: Both … . They differ in … because … .

Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

At what altitude in units ofR is field one quarter of surface value? At what altitude is it one ninth?

Sentence starter: I notice … . This supports … because … .

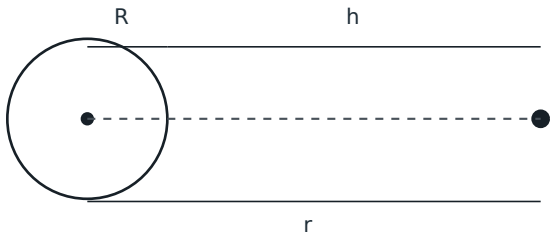
Field / surface field	r / R	h / R
___	___	___
___	___	___

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Find the altitude h/R where a spherical planet's field is 10% of the surface value.
Explain why h/R is not √10.

Sentence starter: My design shows … . It meets the requirement by … .



Marzano check - circle: 1 2 3